Lab - System Utilities

Introduction

In this lab, you will use Windows utilities to configure operating system settings.

Recommended Equipment

- A computer running Windows

Instructions

Part 1: Custom Management Console

Step 1: Explore the management console.

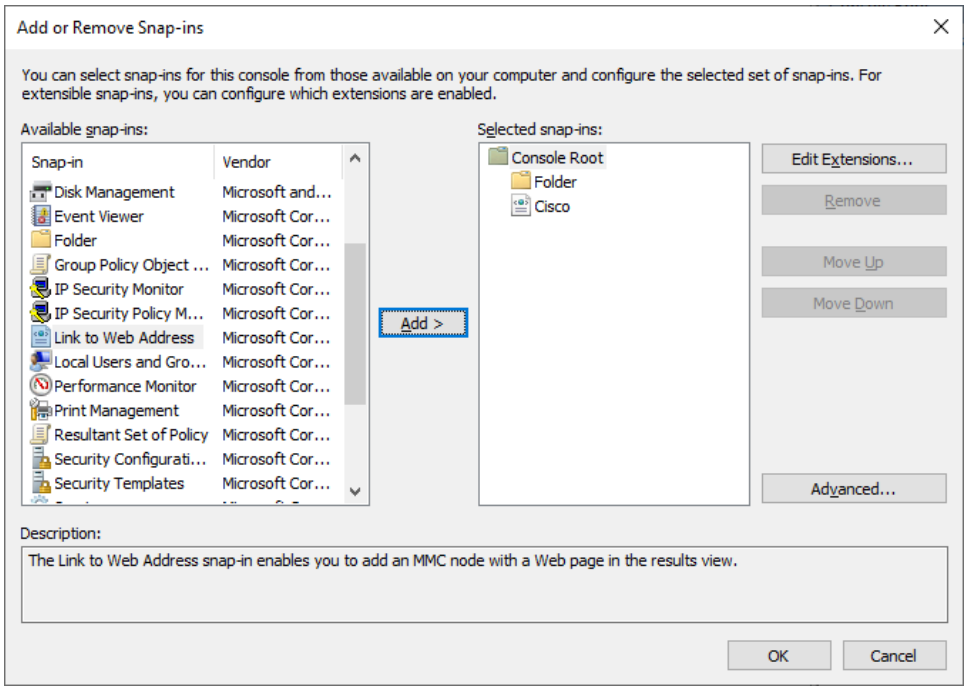
a. To open the Microsoft Management Console, click **Start**, and search for **mmc**. Click **Yes** to allow Microsoft Management Console to make changes to the computer.

b. In the **Console1 - [Console Root]** window, you can build a custom console. Click **File > Add/Remove Snap-in**.

c. In the **Add or Remove Snap-ins** window, you can add a folder snap-in. Scroll down the Available snap-ins: pane to locate the **Folder** snap-in. Select **Folder** and click **Add>**.

d. To add the **Link to Web Address** snap-in, scroll down the Available snap-ins: pane and select **Link to Web Address**. Click **Add>**. In the **Link to Web Address** wizard window, enter http://www.cisco.com in the **Path or URL:** field, click **Next>** to continue.

e. In the Friendly name for the Link to Web Address snap-in box, enter Cisco and click Finish to continue.



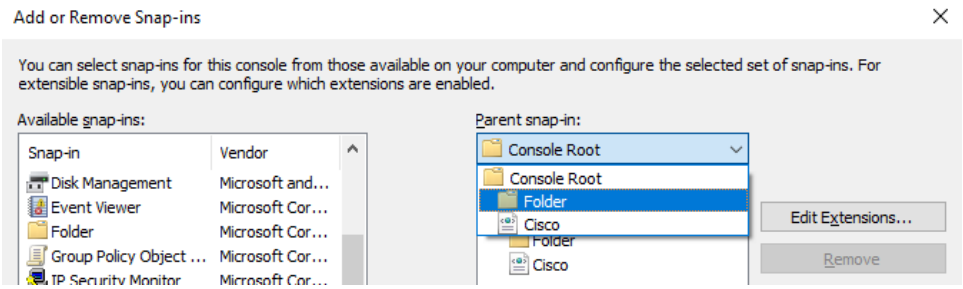
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Step 2: Create a custom management console.

a. To add snap-ins to the folder snap-in, click **Advanced**.

b. The Advanced window opens. Select Allow changing the parent snap-in. Click OK.

c. A drop-down menu appears for **Parent snap-in**. In the **Parent snap-in** box, select **Folder**.



d. Add the following snap-ins from the Available snap-in: pane: **Computer Management**, **Device Manager**, and **Disk Management**. In each of the snap-in windows, click **Finish** to select to use the snap-in to manage the Local Computer or This computer.

e. Click **OK** to continue to exit the Add or Remove Snap-ins window.

f. In the **Console1** window, right-click the **Folder** icon and select **Rename**. Change the name of the folder to **Management Tools**.

g. To save the custom console, click **File > Save As**. Change the file name to your name. Example: **John's Console**. Change the **Save in** box to **Desktop**. Click **Save**.

h. Close all open windows.

i. On the desktop, double-click the **Console** icon that contains your name to re-open the console with your snap-ins. Click **Yes** to allow this application to make changes to your device as necessary.

j. Review the **Management Tools** folder by double-clicking on the tools.

k. Within your custom console, navigate to the Event Viewer.

Question:

What was the path you use to navigate to the Event Viewer?

The path I used to navigate to the Event Viewer was Management Tools > Computer Management > System Tools > Event Viewer.

l. Close the **Console** window when you finish your review. Click **No** when prompted to save the changes.

Part 2: Registry Editor

Step 1: Change your desktop background settings.

a. Right-click the Desktop > select **Personalize**.

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Question:

What is the background picture?

The background picture was the default wallpaper/background that was originally set on my computer. Which is an anime picture.

In Windows 10, select **Solid color** in the Background drop-down list. Choose a **blue** background color.

In Windows 8.1 and 7, click **Desktop Background**. Select **Solid Colors** in the Picture Location drop-down list. Choose a **blue** background color.

b. Click **Save changes**. The computer screen should now have a blue background. If not, ask the instructor for assistance.

c. Close all open windows.

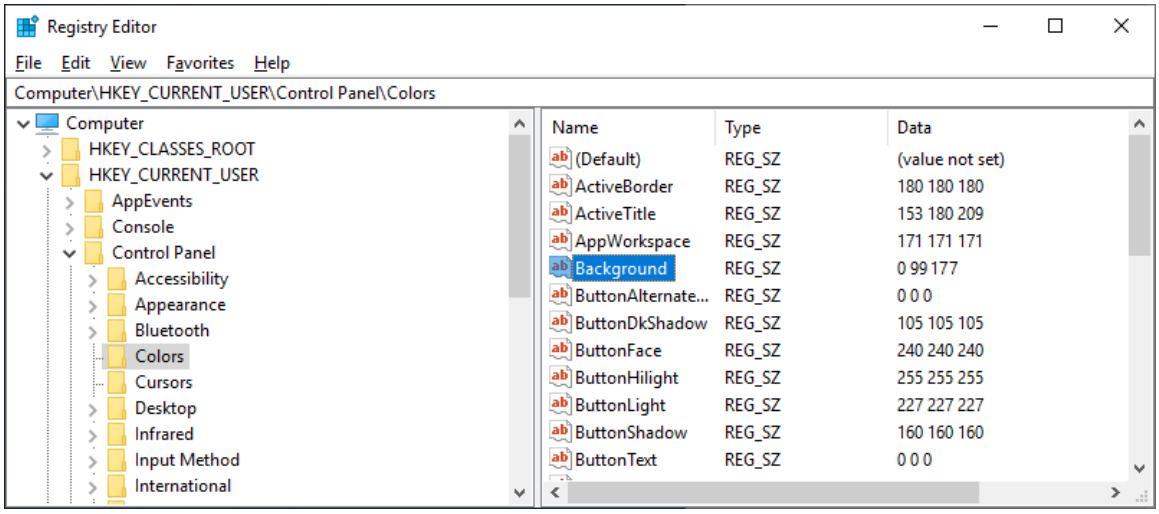
Step 2: Open the registry editor.

a. To open the **Registry Editor**, click **Start** and type **regedit**. Click **Yes** to allow Registry Editor to make changes to the computer.

Note: Do not make any changes in the Registry Editor without instructor permission.

b. Select and expand the **HKEY_CURRENT_USER** entry.

c. Expand **Control Panel**. Click **Colors**. Click **Background** in the right panel.



Question:

What is the data value of the Background (hint – it has three numbers that correspond to red, green, and blue)?

The data value of the Background was 0 0 255, which represents the RGB values for blue.

Step 3: Export a registry key.

You will export the **HKEY_CURRENT_USER\Control Panel\Colors** folder.

a. In the left pane, select the **Colors** folder and click **File > Export**.

b. Save the file to the Desktop with the name **BlueBKG**.

c. At the desktop, right-click the **BlueBKG.reg** icon and select **Edit**.

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d. Notepad opens displaying the contents of **BlueBKG.reg**.

Question:

What is the data value of the **Background**?

The data value of the Background in the BlueBKG.reg file was 0 0 255, which represents the blue desktop background.

e. Close the **BlueBKG.reg – Notepad** window.

f. Open the Personalization window by right-clicking the **Desktop > Personalize > Desktop Background**. Select a **red** background color.

g. Click **Save** changes. Close the **Personalization** window.

h. The desktop background should be red. Click the **Registry Editor** window.

i. On your keyboard, press **F5** to refresh the **Registry Editor** window.

Question:

What is the data value of the Background? Compare the new value in the Registry to the value saved in BlueBKG.reg.

After changing the desktop background to red, the data value of the Background changed to 255 0 0. These numbers represent the RGB values for red.

Step 4: Import a registry file.

You will import the **BlueBKG.reg** file to change the Desktop ground back to blue.

a. Click the **Registry Editor** window.

b. Click **File** > **Import**. Locate and click the **BlueBKG.reg** file then click **Open**.

c. The **Registry Editor** informational message opens letting you know that keys and values have been successfully added to the registry. Click **OK**.

d. Select the **Registry Editor** window.

Questions:

What is the data value of the Background?

After importing the BlueBKG.reg file, the data value of the Background changed back to 0 0 255, which represents blue.

What is the color of the desktop?

After importing the BlueBKG.reg file, the desktop color was blue because the registry file restored the settings that were saved when the background was blue.

e. Restart the computer.

Question:

What is the color of the desktop?

After restarting the computer, the desktop color was blue. The blue background setting was restored through the registry file, so the setting remained after the computer was restarted.

f. Reset **Display Properties Background** to the original settings.

g. Close all open windows.

h. Delete the custom console and BlueBKG file on the desktop.

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Reflection Question

1. Why would it be beneficial to add snap-ins to the mmc that are for other computers on the network?

   It would be beneficial because it allows an administrator to manage and monitor other computers on the network from one location. Instead of going to each computer individually, the administrator can use the MMC snap-ins to check settings, devices, and other system tools remotely. This can save time and make managing multiple computers much easier.

2. After restarting the computer, why was the desktop color changed?

   The desktop color changed after restarting because the blue background setting was restored through the registry. When the BlueBKG.reg file was imported, it changed the Background registry value back to the value that was saved when the desktop was blue. Because the registry stores important Windows settings, the blue background remained even after the computer was restarted.

Help



```
Microsoft Windows [Version 10.0.26200.9457]
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C:\Users\Erika>help
For more information on a specific command, type HELP command-name
ASSOC          Displays or modifies file extension associations.
ATTRIB         Displays or changes file attributes.
BREAK          Sets or clears extended CTRL+C checking.
BCDEDIT        Sets properties in boot database to control boot loading.
CACLS          Displays or modifies access control lists (ACLs) of files.
CALL           Calls one batch program from another.
CD             Displays the name of or changes the current directory.
CHCP           Displays or sets the active code page number.
CHDIR          Displays the name of or changes the current directory.
CHKDSK         Checks a disk and displays a status report.
CHKNTFS        Displays or modifies the checking of disk at boot time.
CLS            Clears the screen.
CMD            Starts a new instance of the Windows command interpreter.
COLOR          Sets the default console foreground and background colors.
COMP           Compares the contents of two files or sets of files.
COMPACT        Displays or alters the compression of files on NTFS partitions.
CONVERT        Converts FAT volumes to NTFS.  You cannot convert the
               current drive.
COPY           Copies one or more files to another location.
DATE           Displays or sets the date.
DEL            Deletes one or more files.
DIR            Displays a list of files and subdirectories in a directory.
DISKPART       Displays or configures Disk Partition properties.
DOSKEY         Edits command lines, recalls Windows commands, and
               creates macros.
DRIVERQUERY    Displays current device driver status and properties.
ECHO           Displays messages, or turns command echoing on or off.
ENDLOCAL       Ends localization of environment changes in a batch file.
ERASE          Deletes one or more files.
EXIT           Quits the CMD.EXE program (command interpreter).
FC             Compares two files or sets of files, and displays the
               differences between them.
FIND           Searches for a text string in a file or files.
FINDSTR        Searches for strings in files.
FOR            Runs a specified command for each file in a set of files.
FORMAT         Formats a disk for use with Windows.
FSUTIL         Displays or configures the file system properties.
FTYPE          Displays or modifies file types used in file extension
               associations.
GOTO           Directs the Windows command interpreter to a labeled line in
               a batch program.
GPRESULT       Displays Group Policy information for machine or user.
HELP           Provides Help information for Windows commands.
ICACLS         Display, modify, backup, or restore ACLs for files and
               directories.
IF             Performs conditional processing in batch programs.
LABEL          Creates, changes, or deletes the volume label of a disk.
MD             Creates a directory.
MKDIR          Creates a directory.
MKLINK         Creates Symbolic Links and Hard Links
MODE           Configures a system device.
MORE           Displays output one screen at a time.
MOVE           Moves one or more files from one directory to another
               directory.
OPENFILES      Displays files opened by remote users for a file share.
PATH           Displays or sets a search path for executable files.
PAUSE          Suspends processing of a batch file and displays a message.
POPD           Restores the previous value of the current directory saved by
               PUSHD.
PRINT          Prints a text file.
PROMPT         Changes the Windows command prompt.
PUSHD          Saves the current directory then changes it.
RD             Removes a directory.
RECOVER        Recovers readable information from a bad or defective disk.
REM            Records comments (remarks) in batch files or CONFIG.SYS.
REN            Renames a file or files.
RENAME         Renames a file or files.
REPLACE        Replaces files.
RMDIR          Removes a directory.
ROBOCOPY       Advanced utility to copy files and directory trees
SET            Displays, sets, or removes Windows environment variables.
SETLOCAL       Begins localization of environment changes in a batch file.
SC             Displays or configures services (background processes).
SCHTASKS       Schedules commands and programs to run on a computer.
SHIFT          Shifts the position of replaceable parameters in batch files.
SHUTDOWN       Allows proper local or remote shutdown of machine.
SORT           Sorts input.
START          Starts a separate window to run a specified program or command.
SUBST          Associates a path with a drive letter.
SYSTEMINFO     Displays machine specific properties and configuration.
TASKLIST       Displays all currently running tasks including services.
TASKKILL       Kill or stop a running process or application.
TIME           Displays or sets the system time.
TITLE          Sets the window title for a CMD.EXE session.
TREE           Graphically displays the directory structure of a drive or
               path.
TYPE           Displays the contents of a text file.
VER            Displays the Windows version.
VERIFY         Tells Windows whether to verify that your files are written
               correctly to a disk.
VOL            Displays a disk volume label and serial number.
XCOPY          Copies files and directory trees.
WMIC           Displays WMI information inside interactive command shell.

For more information on tools see the command-line reference in the online help.
```

Command

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```
C:\Users\Erika>dir /?
Displays a list of files and subdirectories in a directory.

DIR [drive:][path][filename] [/A[[:]attributes]] [/B] [/C] [/D] [/L] [/N]
  [/O[[:]sortorder]] [/P] [/Q] [/R] [/S] [/T[[:]timefield]] [/W] [/X] [/4]

  [drive:][path][filename]
              Specifies drive, directory, and/or files to list.

  /A          Displays files with specified attributes.
  attributes   D  Directories                R  Read-only files
               H  Hidden files               A  Files ready for archiving
               S  System files               I  Not content indexed files
               L  Reparse Points             O  Offline files
               -  Prefix meaning not
  /B          Uses bare format (no heading information or summary).
  /C          Display the thousand separator in file sizes.  This is the
              default.  Use /-C to disable display of separator.
  /D          Same as wide but files are list sorted by column.
  /L          Uses lowercase.
  /N          New long list format where filenames are on the far right.
  /O          List by files in sorted order.
  sortorder    N  By name (alphabetic)       S  By size (smallest first)
               E  By extension (alphabetic)  D  By date/time (oldest first)
               G  Group directories first    -  Prefix to reverse order
  /P          Pauses after each screenful of information.
  /Q          Display the owner of the file.
  /R          Display alternate data streams of the file.
  /S          Displays files in specified directory and all subdirectories.
  /T          Controls which time field displayed or used for sorting
  timefield   C  Creation
              A  Last Access
              W  Last Written
  /W          Uses wide list format.
  /X          This displays the short names generated for non-8dot3 file
              names.  The format is that of /N with the short name inserted
              before the long name. If no short name is present, blanks are
              displayed in its place.
  /4          Displays four-digit years

Switches may be preset in the DIRCMD environment variable.  Override
preset switches by prefixing any switch with - (hyphen)--for example, /-W.
```

Cls

```
C:\Users\Erika>cls /?
Clears the screen.

CLS
```

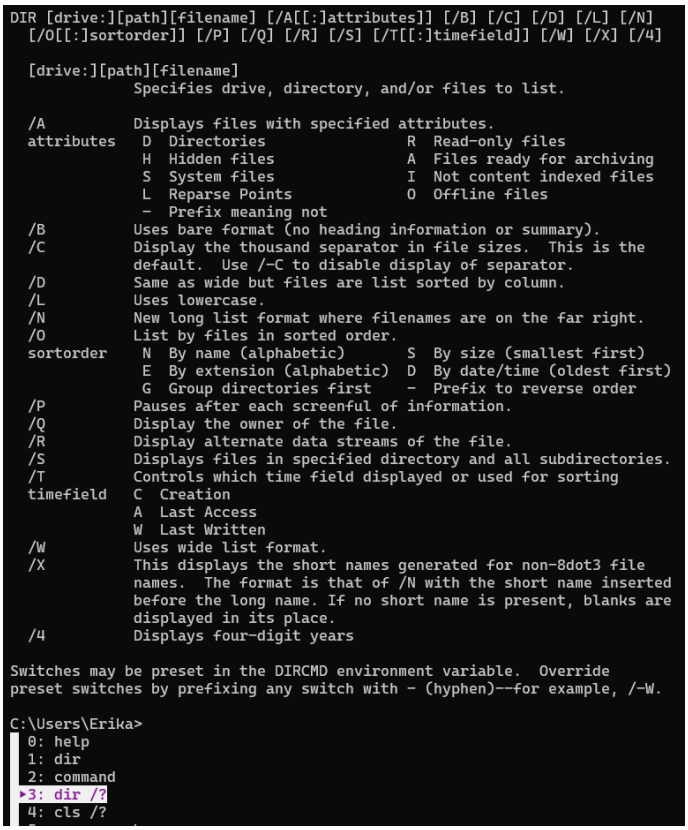
Up-arrow key

```
No command output.
```

F7 Key

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Ctrl+C Keys

```
No command output.
```

Exit

```
No command output.
```

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